

TULIHE, China

WMO: 504340

Lat: 50.483N

Lon: 121.683E

Elev: 733

StdP: 92.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.7	-39.4	-44.7	0.0	-41.3	-42.5	0.1	-39.2	6.4	-20.6	5.6	-20.4	1.2	180	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	28.9	17.6	27.0	17.1	25.0	16.4	19.9	25.3	18.9	23.9	17.9	22.6	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.3	21.9	17.2	13.5	21.1	16.3	12.7	20.1	60.5	25.3	57.0	24.0	53.9	22.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.5	DB	-44.3	32.3	2.3	2.0	-46.0	33.7	-47.3	34.9	-48.6	36.1	-50.2	37.6	(4)
			WB	-44.1	21.9	1.8	1.4	-45.4	22.9	-46.5	23.8	-47.5	24.6	-48.8	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-28.0	-22.8	-12.8	-0.6	7.6	14.1	17.3	15.0	7.7	-2.2	-16.0	-25.6	(6)
(7)		DBStd	16.66	4.61	5.40	6.60	4.99	4.00	3.52	2.60	3.29	4.28	5.26	6.58	5.21	(7)
(8)		HDD10.0	5588	1179	918	706	319	95	7	1	4	95	379	781	1104	(8)
(9)		HDD18.3	8099	1438	1152	964	568	332	132	50	112	321	638	1031	1362	(9)
(10)		CDD10.0	562	0	0	0	1	21	131	227	158	24	0	0	0	(10)
(11)		CDD18.3	31	0	0	0	0	0	6	18	7	0	0	0	0	(11)
(12)		CDH23.3	789	0	0	0	1	44	237	319	166	23	0	0	0	(12)
(13)		CDH26.7	184	0	0	0	0	10	62	75	34	3	0	0	0	(13)
(14)	Wind	WSAvg	2.2	1.6	1.8	2.4	2.9	3.0	2.3	2.1	2.1	2.3	2.3	1.7	1.4	(14)
(15)	Precipitation	PrecAvg	457	4	4	6	21	31	72	132	106	52	14	9	7	(15)
(16)		PrecMax	623	9	9	14	32	62	157	234	237	111	33	15	19	(16)
(17)		PrecMin	295	1	0	1	1	1	17	43	44	5	2	3	2	(17)
(18)		PrecStd	83	2	2	3	8	18	35	45	49	28	9	3	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	-12.2	-3.1	8.5	21.0	28.6	31.8	30.9	30.1	26.5	17.0	3.9	-8.0	(19)	
(20)		MCWB	-13.2	-5.4	1.8	8.5	13.3	17.1	19.1	18.0	14.9	8.2	-0.3	-9.3	(20)	
(21)		DB	-14.9	-6.5	4.9	16.6	24.1	28.4	28.8	27.2	22.9	12.9	-0.2	-12.3	(21)	
(22)		MCWB	-15.7	-8.3	-0.3	6.2	11.7	16.5	18.7	17.9	13.4	5.8	-2.6	-13.1	(22)	
(23)		DB	-17.1	-9.1	1.7	13.0	21.0	26.2	26.8	25.1	20.0	10.0	-3.0	-14.5	(23)	
(24)		MCWB	-17.8	-10.5	-2.2	4.3	10.0	15.7	18.3	17.2	12.3	4.1	-4.6	-15.1	(24)	
(25)		DB	-19.3	-11.7	-0.8	9.6	18.4	23.8	24.8	22.9	17.2	6.8	-5.4	-16.6	(25)	
(26)		MCWB	-19.9	-12.9	-3.9	2.7	8.8	14.8	17.7	16.4	10.8	2.3	-6.7	-17.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	-13.2	-5.4	2.2	9.0	14.3	19.4	21.7	20.6	16.6	8.7	0.2	-9.2	(27)	
(28)		MCDWB	-12.2	-3.5	8.2	20.2	25.6	27.2	27.5	26.6	22.3	15.0	3.1	-8.1	(28)	
(29)		WB	-15.7	-8.2	-0.1	6.6	12.3	18.1	20.4	19.3	14.8	6.3	-2.6	-13.0	(29)	
(30)		MCDWB	-15.0	-6.5	4.4	15.5	21.4	25.3	25.5	24.5	20.2	12.0	-0.3	-12.3	(30)	
(31)		WB	-17.7	-10.5	-2.0	4.8	11.0	16.9	19.5	18.2	13.2	4.5	-4.6	-15.1	(31)	
(32)		MCDWB	-17.1	-9.1	1.6	12.2	19.1	23.1	24.3	22.8	18.4	9.3	-2.9	-14.5	(32)	
(33)		WB	-19.8	-12.8	-3.8	3.1	9.6	15.9	18.6	17.2	11.6	2.7	-6.7	-17.1	(33)	
(34)		MCDWB	-19.3	-11.7	-0.9	9.2	16.9	21.8	23.0	21.5	16.2	6.5	-5.4	-16.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.4	18.6	18.3	15.3	16.5	15.9	13.2	14.0	15.9	14.3	14.5	14.3	(35)	
(36)		5% DB	16.0	19.4	18.1	21.5	22.6	21.1	17.7	18.2	20.9	19.5	15.9	14.2	(36)	
(37)		MCWBR	15.3	17.5	13.3	12.3	11.3	10.0	8.7	9.5	11.7	12.5	13.3	13.4	(37)	
(38)		5% WB	15.9	19.3	17.7	19.4	19.1	17.0	13.7	14.3	17.2	17.7	15.3	14.2	(38)	
(39)		MCWBR	15.2	17.5	13.1	11.4	10.1	8.9	7.5	8.1	9.9	11.6	12.9	13.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.252	0.277	0.327	0.420	0.489	0.462	0.430	0.397	0.350	0.323	0.281	0.262	(40)	
(41)		taud	2.182	2.135	2.050	1.988	1.897	2.084	2.242	2.311	2.386	2.346	2.256	2.127	(41)	
(42)		Ebn at Noon	780	852	862	814	772	800	823	834	841	792	734	690	(42)	
(43)		Edh at Noon	73	101	133	161	185	155	130	116	96	82	68	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.61	2.79	4.31	5.17	5.54	5.90	5.35	4.87	3.88	2.62	1.60	1.22	(44)	
(45)		RadStd	0.06	0.11	0.25	0.31	0.49	0.56	0.44	0.48	0.31	0.20	0.09	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-2.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page